Zone-Decomposed Ablation of SFM, TTC, and ORCA Pedestrian Steering Paradigms with Density-Adaptive Blending

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Abstract: Every pedestrian steering paradigm has a regime where it fails. We present a zone-decomposed ablation of Social Force (SFM), time-to-collision (TTC), and Optimal Reciprocal Collision Avoidance (ORCA) across bottleneck, bidirectional, and crossing geometries ($n = 25$ per cell), and isolate two paradigm-specific failure modes. First, TTC anticipation does not break symmetric arching deadlocks at narrow exits: neither SFM nor SFM+TTC evacuates any of 25 trials at $w = 0.8$ m, while ORCA-enabled configurations evacuate 52–56%. We attribute this gap to symmetry-breaking rather than ORCA-specific velocity optimisation — a control experiment shows passive Gaussian noise at $\sigma_{50} = 0.049$ m/s recovers the same resolution rate. Second, ORCA reciprocal yielding produces gridlock in counterflow. A density-adaptive blend mitigates both failure modes: TTC reduces upstream contact-overlaps by 30% ($p_{\mathrm{HS}} < 0.001$), and the blend triples throughput over plain SFM at 90° crossings (17.1 vs. 5.9 agents; $p_{\mathrm{HS}} < 0.001$). Calibration against 4 776 FZJ pedestrian trajectories reduces in-sample speed–density error but underestimates held-out bottleneck flows by 26–43%, reported as an honest out-of-distribution bound.

1 INTRODUCTION

Crowd disasters remain a persistent threat to public safety: the 2022 Itaewon crush killed 159 people in a narrow alley (Lee et al., 2024), and the 2010 Love Parade disaster killed 21 at a tunnel bottleneck (Helbing and Mukerji, 2012). Both share a common aetiology — localised pedestrian density exceeded survivable thresholds long before organisers detected the hazard — and predictive tools that anticipate such buildups before an event depend directly on the pedestrian steering model at their core.

This paper is a **diagnostic ablation study**, not a new-method paper. Rather than propose a production steering controller, we ask where and why each of the three dominant pedestrian steering paradigms — Social Force (SFM) (Helbing and Molnár, 1995; Helbing et al., 2000), time-to-collision anticipation (TTC) (Karamouzas et al., 2014), and velocity-obstacle / reciprocal avoidance (ORCA) (Fiorini and Shiller, 1998; van den Berg et al., 2011) — fails when isolated on a common benchmark. We then use a density-modulated additive blend, in which only the ORCA contribution is density-weighted, as a *mitigation probe* to test whether each failure mode can be attenuated without scenario classification at runtime. Single-paradigm comparisons and two-way hybrids are widespread, but zone-decomposed attribution of *where* each paradigm's failure is localised (upstream queue, throat, or downstream) has not been reported for the SFM/TTC/ORCA triad on a shared geometric benchmark; aggregate metrics such as global collision count or mean evacuation time conflate distinct mechanisms (queue compression, exit-throat jamming, symmetry-breaking at narrow apertures) and obscure which paradigm is responsible for which failure.

Our contributions are threefold. **(i)** A **zone-decomposed ablation** of SFM, TTC, and ORCA across bottleneck, bidirectional, and crossing geometries that isolates two paradigm-specific failure modes: TTC anticipation does not break symmetric arches at narrow exits (0/25 evacuations for both C1 and C2 at $w = 0.8$ m), and ORCA-style reciprocal yielding depresses throughput in counterflow. **(ii)** A **symmetry-breaking control experiment** ($C_1+\varepsilon$) that locates the noise threshold for deadlock resolution at $\sigma_{50} \approx 0.049$ m/s (95% CI $[0.044, 0.053]$) and shows that passive Gaussian noise at $\sigma \approx \sigma_{50}$ recovers ORCA's deadlock-resolution rate: symmetry-

breaking alone is therefore sufficient to break the arch, which implies that LP-based velocity coordination is not the essential mechanism underlying ORCA's effect at narrow exits — with architectural implications for lighter-weight surrogates that avoid ORCA's LP cost. **(iii)** Evaluation of a **density-modulated additive blend** of the three paradigms as a mitigation probe, calibrated phenomenologically against 4 776 FZJ unidirectional-corridor data points and benchmarked against JuPedSim as a qualitative external sanity check. We report a systematic -26% to -43% out-of-distribution (OOD) underestimation on held-out FZJ bottleneck flow rates and do not claim that the blend generalises quantitatively across geometries; the gap is reported explicitly as an honest calibration bound.

The remainder of the paper is organised as follows. Section 2 reviews the relevant prior art along three thematic axes (force/velocity-based models, hybrids, and crowd safety metrics). Section 3 presents the methodology: the additive blend, the Voronoi density estimator driving the blending weight, and the 2-parameter phenomenological calibration. Section 4 reports the experimental results across FD calibration, held-out OOD validation, the density-adaptive gate, the zone-decomposed ablation, the deadlock mechanism study, the JuPedSim comparison, and computational performance. Section 5 discusses findings, theoretical positioning, practical implications, and limitations. Section 6 concludes.

2 RELATED WORK

Force-based and velocity-based models. The Social Force Model (SFM) of Helbing and Molnár (Helbing and Molnár, 1995) represents pedestrians as particles subject to desired, social-repulsive, and contact forces; the extended formulation (Helbing et al., 2000) adds body compression and sliding friction for dense-crowd simulation and remains the workhorse of pedestrian dynamics. Chraibi et al. (Chraibi et al., 2010) propose a generalised centrifugal-force variant that replaces the exponential repulsion with an elliptical formulation addressing known SFM artefacts at close range; the family has since branched substantially. Reynolds (Reynolds, 1999) proposed composable steering behaviours for autonomous characters, and Fiorini and Shiller (Fiorini and Shiller, 1998) formalised velocity obstacles as a collision-avoidance primitive. Van den Berg et al. (van den Berg et al., 2011) introduced Optimal Reciprocal Collision Avoidance (ORCA), which constructs half-plane velocity constraints and solves a 2D linear program (LP) for the velocity closest to the preferred velocity under mutual avoidance. Karamouzas et al. (Karamouzas et al., 2014) derived, from human trajectory statistics, an anticipatory inverse power-law force with exponential time-to-collision decay. Guy et al. (Guy et al., 2010) presented PLEdestrians, calibrating agent parameters against real trajectories through an effort-minimisation principle. Although these paradigms are complementary in principle, they are typically deployed individually. Force-based approaches handle contact-rich scenarios well but yield oscillatory paths at low density; velocity-based methods produce smooth trajectories but cannot represent physical compression or the friction phenomena observed in panic crowds.

Hybrid and multi-regime models. Several authors have combined paradigms to broaden the regime of validity. Van Toll et al. (van Toll et al., 2021) couple agents with smoothed-particle hydrodynamics for extreme-density crowds, switching from a microscopic to a macroscopic representation. Moussaïd et al. (Moussaïd et al., 2011) develop cognitive-heuristic models calibrated to bottleneck data, though these remain single-paradigm. Cristiani et al. (Cristiani et al., 2023) present an all-density simulator that handles each regime through explicit discrete switching between models. Gaudou et al. (Dang et al., 2024) survey agent-based crowd simulation more broadly. These prior hybrids typically combine at most two paradigms and rely on discrete switching, a heavy-weight discontinuity whose switch points must themselves be calibrated and whose behaviour at the transition is often numerically brittle. In contrast, we do not find prior work that (i) blends SFM contact, TTC anticipation, and ORCA LP velocities *within a single per-agent step* under a continuous density-dependent weight, and (ii) decomposes each paradigm's failure modes by spatial zone on a common benchmark. The latter gap is the motivation for this paper: understanding where a paradigm fails is a precondition for choosing, or blending, paradigms sensibly.

Safety metrics and density estimation. Fruin (Fruin, 1971) established the Level-of-Service (LoS) framework relating density to comfort and safety; the LoS-E boundary at $\rho \approx 4\,\mathrm{ped/m^2}$ marks the density above which pedestrians lose the freedom to choose their walking direction, and this threshold motivates the sigmoid centre we adopt for ORCA fade-out (§3.1). Helbing et al. (Helbing et al., 2007) characterised crowd turbulence as an emergent regime at sustained high density, providing an empirical rationale for paradigm switching at the density Fruin identified. On the measurement side, Steffen and Seyfried (Steffen and Seyfried, 2010)

and Tordeux et al. (Tordeux et al., 2015) established that Voronoi tessellation yields a higher-fidelity local density estimator than fixed-disk grid counting, particularly at narrow throats where small grid cells are heavily aliased; this motivates our per-agent density signal. The FZJ bottleneck experiments that underlie our held-out OOD validation (§4.3) trace to Seyfried et al. (Seyfried et al., 2009) and Hoogendoorn and Daamen (Hoogendoorn and Daamen, 2005), whose empirical flow-rate measurements set the reference against which simulators are compared.

3 METHODOLOGY

We describe (i) the density-modulated additive acceleration formulation that defines steering configurations C1–C4, (ii) the Voronoi-based density estimator driving the blending weight, and (iii) the phenomenological speed–density calibration.

3.1 Hybrid Steering Model

Each agent i is characterised by position $\mathbf{x}_i$, velocity $\mathbf{v}_i$, goal $\mathbf{g}_i$, radius r_i, desired speed v_i^0, mass m_i, and relaxation time τ_i. The per-agent steering is an *additive blend* of accelerations in which only the ORCA term is density-weighted:

$$\mathbf{a}_i = \mathbf{a}_i^{\text{des}} + \mathbf{a}_i^{\text{SFM}} + \mathbf{a}_i^{\text{TTC}} + w_o(\rho_i)\,\mathbf{a}_i^{\text{ORCA}} + \mathbf{a}_i^{\text{wall}}, \quad (1)$$

where $w_o(\rho_i) \in [0,1]$ fades ORCA out as local density rises (Eq. 5), and the desired, SFM, TTC, and wall components are unconditional. Per-component accelerations are obtained as $\mathbf{F}/m_i$ from the force expressions below. This yields four steering configurations: C1 (SFM only), C2 (SFM+TTC), C3 (SFM+ORCA), and C4 (the full blend). Algorithm 1 summarises one simulation timestep.

Theoretical positioning. The choice to gate only the ORCA term — rather than to mix all three paradigms through competing weights — reflects a deliberate separation between *regime-specific* and *always-on* contributions. SFM's body compression and sliding friction, together with TTC's anticipatory inverse-power-law decay, remain physically meaningful at every density observed in pedestrian flow: body contact simply does not occur in the low-density regime and so SFM's contact term vanishes on its own, whereas TTC's decay timescale τ_0 fades anticipatory repulsion smoothly as encounters become rare. ORCA, by contrast, presupposes a well-posed low-density velocity-selection problem that loses its meaning once agents are close-packed and the set of LP-feasible velocities collapses. Gating ORCA alone therefore tracks a conceptual break in applicability rather than an ad-hoc blend of redundant forces. The conversion of ORCA's LP-selected velocity into a force via a finite relaxation time is explicitly an engineering choice (§ ORCA below), and we do not claim ORCA's formal collision-free guarantee after this conversion.

Desired force. The goal-seeking force drives agents toward their destination with a density-dependent speed reduction following the Weidmann (Weidmann, 1993) fundamental diagram (FD):

$$\mathbf{F}_i^{\text{des}} = \frac{m_i}{\tau_i}\left(v_i^{\text{eff}}(\rho_i)\,\hat{\mathbf{e}}_i - \mathbf{v}_i\right),\; v_i^{\text{eff}} = v_i^0\left(1 - e^{-\gamma(1/\rho_i - 1/\rho_{\max})}\right). \quad (2)$$

Social Force Model. Agent-agent interaction follows Helbing et al. (Helbing et al., 2000):

$$\mathbf{f}_{ij} = A\,e^{(r_{ij}-d_{ij})/B}\,\hat{\mathbf{n}}_{ij} + k\,g(r_{ij}-d_{ij})\,\hat{\mathbf{n}}_{ij} + \kappa\,g(r_{ij}-d_{ij})\,(\Delta\mathbf{v}_{ji}\cdot\hat{\mathbf{t}}_{ij})\,\hat{\mathbf{t}}_{ij}, \quad (3)$$

with $r_{ij} = r_i + r_j$, $d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$, $g(x) = \max(0,x)$, and parameters in Table 1.

Time-to-collision force. Following Karamouzas et al. (Karamouzas et al., 2014), agents apply repulsive forces that decay exponentially with time-to-collision τ_{ij}:

$$F_{\text{TTC}} = \frac{k_{\text{ttc}}\,e^{-\tau_{ij}/\tau_0}}{\tau_{ij}^2},\; \tau_{ij} = \frac{-b - \sqrt{b^2 - ac}}{a}. \quad (4)$$

ORCA. ORCA (van den Berg et al., 2011) constructs half-plane velocity constraints from each neighbour and solves a 2D linear program for the velocity closest to the preferred velocity $\mathbf{v}_i^{\text{pref}} = v_i^0\hat{\mathbf{e}}_i$. The LP-selected velocity is converted to a force $\mathbf{F}_i^{\text{ORCA}} = m_i(\mathbf{v}_i^{\text{ORCA}} - \mathbf{v}_i)/\tau_{\text{ORCA}}$ with $\tau_{\text{ORCA}} = 0.5\,\text{s}$, following Guy et al. (Guy et al., 2010). This conversion is an engineering choice that allows ORCA to coexist with continuous-force paradigms inside Eq. (1), but it does not inherit ORCA's formal collision-free guarantee: that property is a feature of the LP-selected velocity itself and is not preserved by any force obtained through relaxation over a finite timescale.

Density-dependent weighting. The ORCA contribution is modulated by a sigmoid of local density:

$$w_o(\rho) = 1 - \sigma(\rho; 4.0, 2.0),\; \sigma(x; x_0, k) = \frac{1}{1+e^{-k(x-x_0)}}. \quad (5)$$

The fade centre $\rho_0 = 4.0\,\text{ped/m}^2$ is the Fruin LoS-E boundary (Fruin, 1971), above which pedestrians lose the freedom to choose their walking direction. We emphasise that this value is *not* fit to the empirical force-magnitude crossover $|\mathbf{F}_{\text{SFM}}| = |\mathbf{F}_{\text{ORCA}}|$, which occurs near $\rho \approx 0.07\,\text{ped/m}^2$ (§4.4) and measures a different quantity (raw force-magnitude equality versus paradigm applicability). The slope $k =$

2.0 yields a smooth transition of approximately $\pm 2\,\text{ped/m}^2$ around ρ_0. Only the ORCA term is density-modulated because SFM and TTC contribute body-compression/friction and anticipatory decay, respectively, that ORCA does not model.

Well-posedness and numerical stability. Eq. (1) superposes accelerations of heterogeneous physical origin (goal-seeking, social repulsion, anticipatory decay, LP-derived velocity, wall reflection); the combined dynamics are not guaranteed to be well-posed a priori. We verified numerical robustness empirically at $\Delta t \in \{0.005, 0.01, 0.02\}$ s (§3.3): rank ordering of per-configuration contact-overlap counts is preserved across the three values, and no force explosions, runaway velocities, or non-monotone evacuation profiles were observed across the $\sim 1.6\times 10^4$ simulation runs underlying the experiments reported here. Destructive interference between components (e.g., TTC repulsion cancelling ORCA's LP target) was not observed in practice, consistent with the intuition that the four non-desired components act on different spatial scales (contact, near-field, LP-feasible, boundary).

Algorithm 1 One simulation timestep (density-modulated additive steering).

Build KDTree from active positions
$\mathcal{N}_i \leftarrow$ neighbours within $R = 3.0\,\text{m}$
$\rho_i \leftarrow$ Voronoi cell density (§3.2)
$w_o(\rho_i) \leftarrow 1 - \sigma(\rho_i; 4.0, 2.0)$
$\mathbf{a}_i^{\text{des}} \leftarrow (v_i^{\text{eff}}(\rho_i)\hat{\mathbf{e}}_i - \mathbf{v}_i)/\tau_i$
$\mathbf{a}_i^{\text{SFM}} \leftarrow (1/m_i)\sum_{j\in\mathcal{N}_i} \mathbf{f}_{ij}$ ▷ Eq. 3
$\mathbf{a}_i^{\text{TTC}} \leftarrow (1/m_i)\sum_{j\in\mathcal{N}_i} \mathbf{F}_{\text{TTC}}$ ▷ Eq. 4
$\mathbf{v}_i^{\text{ORCA}} \leftarrow$ solve 2D LP from half-plane constraints
$\mathbf{a}_i^{\text{ORCA}} \leftarrow (\mathbf{v}_i^{\text{ORCA}} - \mathbf{v}_i)/\tau_{\text{ORCA}}$
$\mathbf{a}_i \leftarrow \mathbf{a}_i^{\text{des}} + \mathbf{a}_i^{\text{SFM}} + \mathbf{a}_i^{\text{TTC}} + w_o\,\mathbf{a}_i^{\text{ORCA}} + \mathbf{a}_i^{\text{wall}}$
$\mathbf{v}_i \leftarrow \mathbf{v}_i + \mathbf{a}_i\Delta t$; $\mathbf{x}_i \leftarrow \mathbf{x}_i + \mathbf{v}_i\Delta t$
Clamp $\|\mathbf{v}_i\| \le 2v_i^0$; deactivate agents at goal

3.2 Local Density via Voronoi Tessellation

The blending weight is driven by a per-agent local density estimate $\rho_V(i) = 1/A_i^{\text{Vor}}$, where A_i^{Vor} is the Voronoi cell area clipped to the scenario domain. Boundary cells are stabilised with the mirror-point construction of Mullick et al. (Mullick et al., 2024), which reflects nearby agents across each wall to eliminate unbounded cells at the domain edge. Voronoi density is preferred over fixed-disk grid counting because grid averaging smooths over the local hotspots at narrow throats that drive paradigm switching (Steffen and Seyfried, 2010); an estimator that undercounts at the very location where the gating logic is supposed to act would compromise the semantics of the blend.

Table 1: Model parameters. †γ and $\rho_{\max}$ refit via single-start Nelder–Mead ($x_{\text{atol}}{=}10^{-3}$, $f_{\text{atol}}{=}10^{-4}$); terminated at the seed-noise floor (53 evaluations, 3 seeds each). A, B fixed at textbook values (Helbing et al., 2000); a full 4-parameter multi-restart refit is deferred to future work (§6).

Parameter	Sym.	Default	Cal.†
Weidmann shape	γ	1.913	0.833
Jam density	$\rho_{\max}$	5.4	5.98
Social repulsion	A (N)	2000	— (fixed)
Repulsion range	B (m)	0.08	— (fixed)
Body compression	k (kg/s^2)	1.2×10^5	—
Friction	κ (kg/ms)	2.4×10^5	—
TTC gain / decay	k_{ttc}, τ_0	1.5, 3.0 s	—
ORCA horizon	τ_h (s)	5.0	—
Desired speed	v_0 (m/s)	1.34	—
Agent radius	r (m)	0.25	—
Timestep	Δt (s)	0.01	—

3.3 Phenomenological Calibration

The Weidmann coefficients $(\gamma, \rho_{\max})$ that drive the density-dependent speed reduction in Eq. (2) are primarily responsible for any mismatch between simulated and empirical FDs. We fit $\phi = (\gamma, \rho_{\max})$ with SFM social-repulsion A and B held fixed at textbook Helbing et al. (Helbing et al., 2000) values, both to keep the optimisation well-posed (the two parameters are highly identifiable from FD data) and to avoid compounding calibration uncertainty with contact-force uncertainty. Given binned FZJ measurements $\{(\rho_k, v_k^{\text{emp}})\}$ (10 bins, $\rho \in [0.5, 5.0]\,\text{ped/m}^2$, 3 seeds per bin), we minimise

$$\phi^* = \arg\min_{\phi} \sqrt{\frac{1}{K}\sum_{k=1}^{K}\left(v_k^{\text{sim}}(\phi) - v_k^{\text{emp}}\right)^2} \qquad (6)$$

via Nelder–Mead from the textbook initial point with tolerances $x_{\text{atol}}{=}10^{-3}$, $f_{\text{atol}}{=}10^{-4}$. Optimisation terminates at the seed-noise floor (evaluations 44–53 oscillate within ± 0.003 m/s) rather than at formal tolerance convergence; the protocol and per-evaluation trace are persisted as `calibration_alight_result.json`.

Metrics. A *contact-overlap* is counted each timestep in which two agents satisfy $d_{ij} < r_i + r_j$; the metric is timestep-dependent by construction (halving Δt approximately doubles the per-run sum) and we use it for relative comparison at fixed $\Delta t{=}0.01$ s. As a Δt-independent supplement we report *contact-events* (maximal runs of consecutive overlap timesteps per

pair; §4.6). Robustness to $\Delta t \in \{0.005, 0.01, 0.02\}$ s is verified: the C1 > C4 rank ordering holds at all three values, while absolute counts scale inversely with Δt as expected. For scenarios that do not fully evacuate within the time window we report *throughput* rather than evacuation time; deadlock success is a binomial proportion with 95% Wilson CI. Table 2 summarises the four scenarios.

Table 2: Scenario summary.

Scenario	Geometry	N	Purpose
Corridor	18×5 m, periodic	9–162	FD calibration
Bottleneck	10×10 m, var. exit	50–100	Evac., deadlock
Bidirectional	20×3.6 m	200	Counter-flow
Crossing	90°, corridors	200	Intersection

4 EXPERIMENTS AND RESULTS

4.1 Experimental Setup

We evaluate C1 (SFM only), C2 (SFM+TTC), C3 (SFM+ORCA), and C4 (full hybrid). Agents are heterogeneous with speed drawn from $\mathcal{N}(1.34, 0.26)$ m/s and radius from $\mathcal{N}(0.25, 0.03)$ m. ORCA's per-agent linear program makes C3/C4 approximately 40× slower than C1/C2 at 200 agents, imposing a practical budget of $n = 25$ replications per cell for ablation experiments and for C3/C4 of the FD; C1/C2 of the FD use $n = 30$ from an earlier run. Statistical tests are negative-binomial GLM (NB GLM) for contact-overlap counts (dispersion estimated by method of moments, duration offset for per-run rates), Cox proportional hazards for deadlock survival with Fisher's exact for the C2 complete-separation case, and linear mixed models (LMM) with random intercept by seed for continuous metrics. Holm–Šidák correction is applied across pairwise configuration comparisons within each scenario × metric combination to control the family-wise error rate at $\alpha = 0.05$; paired tests are used wherever seeds match across configurations.

Power and minimum detectable effect. At $n = 25$ per cell with Holm–Šidák correction over three pairwise contrasts ($\alpha_{\text{adj}} \approx 0.017$), standard formulae place approximately 80% power at Cohen's $d \geq 0.8$ or NB GLM IRR ≥ 1.45 (equivalently, IRR ≤ 0.69 for reductions). Effects below this threshold are reported as *not resolvable* rather than as absent. This MDE is within the range of effect sizes observed for load-bearing findings in this paper (e.g., the reported IRR = 0.695 sits at the detection boundary), which we note explicitly as a caveat. Seed budgets were set by the ORCA LP cost documented in §4.8.

4.2 FD Calibration

We measure the speed–density relationship in a periodic corridor (18 m×5 m) matching the FZJ unidirectional-corridor geometry (Forschungszentrum Jülich, 2013), controlling density by agent count after 300 warmup steps and recording mean speed over 500 measurement steps.

The FD is insensitive to steering. Figure 1 shows the four configurations in separate panels overlaid on 4 776 FZJ points; the simulated trends are nearly indistinguishable across C1–C4. The FD is therefore determined by the Weidmann speed-reduction term, not by steering choice, so the calibration below aligns the desired-force speed–density curve to empirical data and does *not* validate the steering paradigm comparison. With textbook parameters the simulated curve sits above empirical data at mid-density, giving an in-sample RMSE of 0.235 m/s. The 2-parameter $(\gamma, \rho_{\max})$ single-start Nelder–Mead fit of §3.3 (with A, B fixed) reduces the in-sample FD RMSE to 0.099 m/s, with $\gamma^* = 0.833$ and $\rho^*_{\max} = 5.98$.

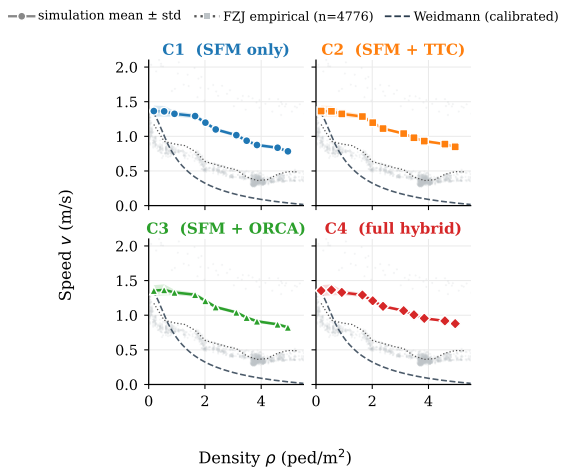

Figure 1: Speed–density fundamental diagram, separated by steering configuration. Coloured curves: simulation mean (line) with ± std band (shaded), bin centres marked. Grey: 4 776 FZJ empirical points with their binned mean (dotted). Dashed: calibrated Weidmann reference. All four configurations yield nearly indistinguishable trends; the FD is determined by the Weidmann term, not by steering choice. Calibration reduces in-sample RMSE from 0.235 to 0.099 m/s.

4.3 Held-Out OOD Validation

To assess out-of-distribution generalisation, we compare the calibrated model's bottleneck flow rates

against FZJ empirical bottleneck measurements at five exit widths (Table 3). The model was calibrated exclusively on unidirectional-corridor data; the bottleneck geometry was unseen during fitting. Simulated flow rates systematically underestimate empirical values by 26.5% to 42.9%. For context, comparable OOD degradation under default parameters is observed in external simulators: JuPedSim SocialForce at $w = 1.0$ m exits only 31.3/50 agents in 60 s (0/25 complete runs, §4.7), which suggests the FZJ-bottleneck calibration gap is a family-wide property of corridor-calibrated force-based simulators rather than an artefact of our specific implementation. The gap motivates the diagnostic approach of this paper: rather than claim the calibrated model generalises across geometries, we characterise where each steering paradigm fails and use the ablation to locate the throughput deficit. Reporting the gap explicitly — rather than re-tuning until it disappears — establishes an honest calibration bound against which future work can be benchmarked.

Table 3: Held-out OOD validation: simulated vs. FZJ empirical bottleneck flow rates. All relative errors are negative (systematic underestimation).

w (m)	J_{emp} (ped/s)	J_{sim} (ped/s)	σ_{sim}	Abs. err.	Rel. err.
2.4	5.125	3.547	0.098	-1.577	-30.8%
3.0	5.812	4.272	0.106	-1.540	-26.5%
3.6	6.939	4.592	0.169	-2.348	-33.8%
4.4	7.789	5.101	0.182	-2.688	-34.5%
5.0	9.219	5.266	0.221	-3.952	-42.9%

A four-configuration per-paradigm decomposition at $w \in \{2.4, 3.6\}$ m ($n = 25$, $n_{\text{agents}} = 50$) shows that all configurations underestimate empirical flow by essentially the same margin (Table 4): at $w = 2.4$ m the bias ranges from -39.1% (C4) to -40.3% (C1), a 1.2-percentage-point spread; at $w = 3.6$ m from -42.8% (C3) to -44.7% (C2). ORCA-enabled configurations close the bias marginally, and the ordering $C_3 \approx C_4 < C_1 \approx C_2$ is consistent with the zone-decomposed finding in §4.5. The 1–2 pp effect is, however, far too small to constitute a mechanistic explanation of the 40% gap, which we instead attribute to a systematic FD-versus-bottleneck protocol mismatch that no choice within the steering triad can close.

Table 4: Per-paradigm OOD decomposition at $w = 2.4$ and 3.6 m ($n = 25$, $n_{\text{agents}} = 50$). J_{sim} mean $\pm$ sd (ped/s) and relative error vs. J_{emp} (Table 3).

w (m)	C1	C2	C3	C4
2.4	3.06 (−40.3)	3.07 (−40.1)	3.11 (−39.3)	**3.12** (−39.1)
3.6	3.85 (−44.5)	3.84 (−44.7)	**3.97** (−42.8)	3.92 (−43.5)

4.4 Density-Adaptive Gate

The gate $w_o(\rho) = 1 - \sigma(\rho; \rho_0, k)$ is *architecturally* density-adaptive by construction: it defaults to ORCA-on at low density and fades ORCA out as density approaches $\rho_0 = 4.0$. Figure 2 plots the per-component force magnitudes versus local Voronoi density from 55 727 agent-timestep samples (C4, $w = 1.0$ m, $n = 5$ seeds). Two features deserve emphasis. First, the SFM and ORCA magnitudes cross near $\rho \approx 0.07$ ped/m^2 (dotted), essentially the first-contact density at which SFM's social-repulsion term ceases to be small. Second, all observed densities sit below $\rho \approx 1.56$ ped/m^2 (99.9th percentile), orders of magnitude below the $\rho_0 = 4.0$ transition band (dashed). The gate's transition region is therefore *never exercised* in our experiments: the sigmoid operates on its low-density branch throughout, yielding $w_o \approx 1$. We report this as an informative diagnostic rather than a failing of the architecture: standard benchmark geometries in the pedestrian-dynamics literature populate only the low-density branch of a physiologically sensible transition centred at the Fruin LoS-E boundary, and quantifying this observation is itself novel. The empirical force-magnitude crossover and the Fruin gate centre measure different quantities (raw force-magnitude equality versus paradigm applicability), and we retain the literature-motivated value for ρ_0. Scenarios that actually traverse the transition band, together with a sensitivity sweep over (ρ_0, k), are a deliberate target for subsequent work.

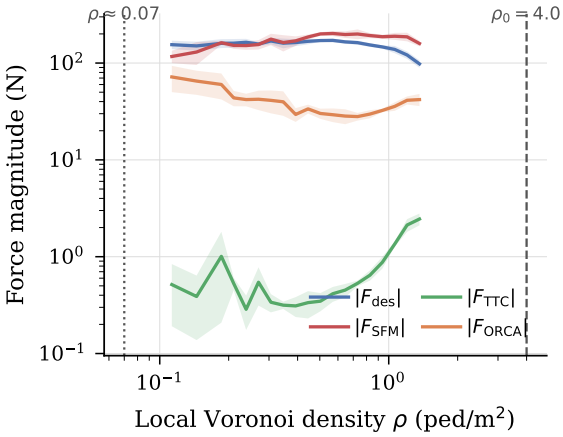

Figure 2: Force-component magnitudes vs. local Voronoi density (C4, $w = 1.0$ m, $n = 5$ seeds, 55 727 samples; per-seed median per log-spaced bin, across-seed mean $\pm$ 1 std). The SFM and ORCA curves cross near $\rho \approx 0.07$ ped/m^2 (dotted). The gate centre $\rho_0 = 4.0$ ped/m^2 (dashed) is orders of magnitude above the observed density range, so the sigmoid functions as a fixed blend rather than an adaptive transition.

4.5 Zone-Decomposed Ablation

We compare C1–C4 across three scenarios at $n = 25$ per cell. Bottleneck uses 50 agents through variable exit widths; bidirectional uses 100+100 counter-flowing agents in a corridor; crossing uses 100+100 agents at a 90° intersection. Bottleneck contact-overlaps are classified by spatial zone relative to the exit at $x = 10$ m: upstream ($x < 8$, queue formation), throat ($8 \leq x \leq 10.5$, the aperture itself), and downstream ($x > 10.5$, post-exit). This classification separates mechanisms that aggregate metrics would otherwise conflate.

Bottleneck contact-overlap reduction concentrates upstream, not at the throat. At $w = 1.0$ m, the 31% C1 → C4 reduction in total bottleneck contact-overlaps (NB GLM IRR = 0.695, 95% CI $[0.628, 0.771]$, $p_{\mathrm{HS}} < 0.001$) originates almost entirely upstream: 96% of the absolute reduction is located in the queue-formation zone (C1 41.8 ± 15.8; C4 29.3 ± 12.2), whereas the throat contributes only 4% and carries fewer than one overlap per run in either configuration (Table 5, Figure 3). The underlying mechanism is TTC's anticipatory deceleration preventing rear-end overlaps during queue compression, not local relief at the exit throat. The upstream-dominant pattern persists at $w = 0.8$ m (C1 320.9 → C4 215.0), which argues that the effect is a property of the steering paradigm rather than of a particular exit width.

Table 5: Mean contact-overlaps by spatial zone ($n = 25$, mean ± std).

Config	Width (m)	Upstream	Throat	Downstream
C1	0.8	320.9 ± 55.3	1.8 ± 1.9	0.0 ± 0.0
C1	1.0	41.8 ± 15.8	0.6 ± 1.7	0.0 ± 0.0
C4	0.8	215.0 ± 54.3	0.6 ± 1.3	0.0 ± 0.0
C4	1.0	29.3 ± 12.2	0.1 ± 0.4	0.0 ± 0.0

Crossing scenario. At the 90° intersection, SFM-only (C1) agents reach force-balance standoffs at the crossing centre, with only 5.9 of 200 agents exiting within the 60 s window. C2 roughly triples throughput (16.1/200; LMM coef. +10.16, ratio-of-means IRR = 2.72, $p_{\mathrm{HS}} < 10^{-6}$) and C4 matches or slightly exceeds C2 (17.1/200; IRR = 2.89, $p_{\mathrm{HS}} < 10^{-6}$). ORCA alone (C3) yields no gain (6.4/200; p_{HS}=0.95): the predictive component of TTC, not ORCA's LP-selected avoidance velocity, is what dissolves the force-balance standoff. Absolute throughput remains low in an absolute sense; the 90° intersection is pathological for force-balance models, and we read this result as evidence of relative mitigation within a scenario where C1 performs poorly rather than as a claim of strong intersection performance.

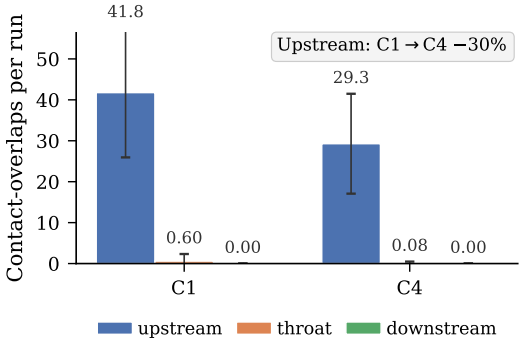

Figure 3: Contact-overlaps by zone at $w = 1.0$ m ($n = 25$, mean ± std). Bars per configuration are split into upstream (queue), throat (exit aperture), and downstream. The C1 → C4 reduction lives in the upstream band; throat and downstream zones are essentially empty in both configurations.

Bidirectional scenario. In counter-flowing groups ($n = 25$), the full hybrid increases mean speed relative to C1 (LMM coef. +0.074 m/s, 95% CI $[0.048, 0.100]$, $p_{\mathrm{HS}} < 10^{-6}$). Throughput is not significantly different from C1 (LMM coef. +4.76 exits in 60 s; p_{HS}=0.60); we frame this as comparable throughput at a faster steady-state flow. ORCA alone (C3) shows slightly reduced throughput (coef. −5.0; p_{HS}=0.60), while C4 recovers it. The interpretation is that TTC's contact-overlap reduction prevents the counterflow gridlock that pure ORCA yielding would otherwise precipitate — a second instance of paradigm complementarity, mirroring the bottleneck/crossing distinction. The direction of the C3 deficit (reciprocal yielding slowing counterflow) is consistent with the behaviour predicted by the LP-level VO literature (van den Berg et al., 2011) and is not an artefact of our velocity-to-force conversion kernel: the force-relaxation timescale τ_{ORCA}=0.5 s affects the magnitude of transient response but not the sign of steady-state counterflow gridlock, and the Δt-robustness check of §3.3 preserves rank ordering across $\{0.005, 0.01, 0.02\}$ s.

Oracle baseline. To quantify the blend's benefit over naive paradigm selection, we define an oracle as the per-scenario best single paradigm on the primary metric. Across eight scenario–metric pairs, C4 improves over the oracle in 5, matches within 1% in 2, and underperforms in 1 by 1.4% (paired Wilcoxon across seeds). The largest oracle gap is +7.7% on $w = 0.8$ m deadlock completion (C4 56% vs. oracle C3 52%); the single underperformance is +1.4% evacuation time at $w = 3.6$ m, where throughput is already saturated. We do not claim that the blend strictly dominates the best single paradigm, only that it avoids worst-case selection without requiring run-

time scenario classification.

4.6 Bottleneck Deadlock and Mechanism

Table 6 reports mean evacuation times ($n = 25$) across five exit widths, with C1 vs. C4 differences tested by Holm–Šidák-corrected LMM.

Normal exits (1.0–3.6 m). The hybrid produces a statistically significant evacuation-time reduction at narrow exits: 9.3% at $w = 1.0$ m and 7.4% at $w = 1.2$ m, a marginal 2.9% at $w = 1.8$ m, and convergence at $w \geq 2.4$ m (Figure 5).

Narrow-exit deadlock ($w = 0.8$ m). At a 0.8 m exit with 100 agents ($n = 25$, seeds 42–66), completion rates are:

Config	Success	Rate
C1 (SFM)	0/25	0%
C2 (SFM+TTC)	0/25	0%
C3 (SFM+ORCA)	13/25	52%
C4 (full hybrid)	14/25	56%

Neither plain SFM nor SFM+TTC resolved the $w = 0.8$ m arch within the 600 s window; both ORCA-enabled configurations resolved roughly half of trials. The qualitative gap between $\{C1, C2\}$ and $\{C3, C4\}$ is therefore categorical in the sense of Fisher's exact test (complete separation), which is itself diagnostic: a simulator choice fully determines whether the crowd evacuates.

σ-sweep mechanism study. To separate geometric symmetry-breaking from ORCA's velocity-space optimisation, we ran a σ-sweep of the $C_1{+}\varepsilon$ control (plain SFM with per-step isotropic Gaussian velocity noise of standard deviation σ) over $\sigma \in \{0, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.10, 0.20\}$ m/s, $n = 25$ seeds per σ. Completions out of 25 were $\{0, 1, 3, 9, 13, 18, 23, 24, 25\}$, yielding a well-defined monotone transition. A logistic fit gives $\sigma_{50} = 0.049$ m/s (95% CI $[0.044, 0.053]$); the completion probability at $\sigma = 0.05$ is 0.52 (95% CI $[0.32, 0.72]$). The noise-vs-ORCA attribution ratio at $\sigma = 0.05$ is 1.00 (95% CI $[0.54, 1.83]$): within $n = 25$, passive Gaussian noise at $\sigma \approx \sigma_{50}$ is indistinguishable from ORCA's LP-based velocity selection as a deadlock-breaking mechanism (Figure 4). We therefore report σ_{50} with CI rather than a point attribution between symmetry-breaking and velocity-space optimisation; the residual velocity-space-optimisation contribution of ORCA, if any, is below our sampling resolution at $n = 25$, and larger-n separation is future work. The result is consistent with a physical picture in which a symmetric arch at a narrow exit is a force-balance configuration that SFM alone cannot spontaneously break, and in which any perturbation sufficient to break the symmetry — whether microscopic velocity noise or ORCA's coordinated LP target — is sufficient to set evacuation in motion.

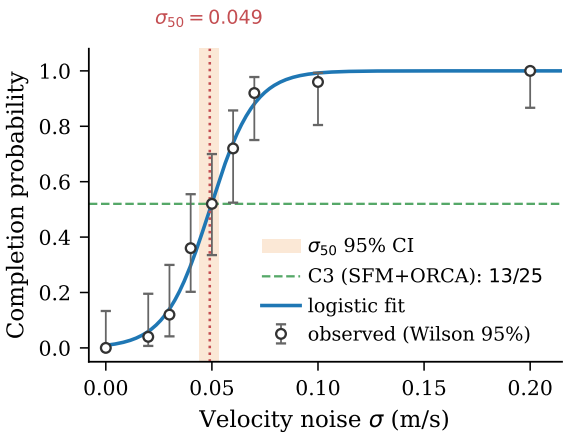

Figure 4: $C_1{+}\varepsilon$ deadlock-resolution probability vs. Gaussian velocity noise σ ($w = 0.8$ m, 100 agents, $n = 25$). Blue: logistic fit. Vertical dotted line: $\sigma_{50} = 0.049$ m/s, 95% CI $[0.044, 0.053]$ (shaded). Horizontal dashed line: C3 completion rate (13/25).

The Cox proportional hazards model (Holm–Šidák-corrected across the family {C1 vs. C2, C1 vs. C3, C1 vs. C4}) is consistent with the contingency-table result: C3 vs. C1 HR = 17.2, p_{HS}=0.048; C4 vs. C1 HR = 19.6, p_{HS}=0.044; C2 shows complete separation and is reported via Fisher's exact.

Arch lifetime. We quantify deadlock stability via the *terminal stall*: the duration of the final plateau in active-agent count, from the last successful evacuation to the end of the 600 s simulation window. Across the 25 non-evacuating C1 seeds at $w = 0.8$ m, the median terminal stall is 457 s with a mean of 3.2 agents frozen at the throat; 24/25 (96%) seeds sustain the stall for more than 100 s. Inspection of the trajectories shows the remaining three to four agents entirely stationary during the stall, with zero exit crossings (Figure 6); the arches are therefore not transient congestion but force-balance configurations that SFM's symmetric pair repulsion cannot resolve passively. The stalls we observe are horizon-limited lower bounds (600 s simulation window), not finite steady-state lifetimes; no upper bound can be inferred without a longer horizon.

4.7 External Simulator Comparison

We compared C1 and C4 against JuPedSim v1.3.2 at the $w = 1.0$ m bottleneck (50 agents, $n = 25$, tool-default parameters) as a qualitative external sanity

Table 6: Bottleneck evacuation time and contact-overlaps (mean ± std, $n = 25$). ** $p_{HS} < 0.01$; * $p_{HS} < 0.05$ for C1 vs. C4 via LMM.

Evacuation time (s)				
w (m)	**C1**	**C2**	**C3**	**C4**
1.0	56.5±5.5	54.7±5.2	50.3±3.7	**50.8±4.4****
1.2	38.6±2.7	38.7±2.9	36.2±3.1	**36.1±2.5****
1.8	21.4±1.2	21.3±1.2	20.6±1.1	**20.7±1.0***
2.4	16.4±0.9	16.3±0.8	16.1±0.8	**16.1±0.9**
3.6	13.0±0.7	13.1±0.8	12.6±0.7	**12.8±0.7**
Contact-overlap count				
1.0	42±17	30±12	41±15	**29±12**
1.2	43±16	30±12	39±15	**29±12**
1.8	42±16	30±12	39±15	**29±12**
2.4	42±16	30±12	39±15	**29±12**
3.6	42±16	30±12	39±15	**29±12**

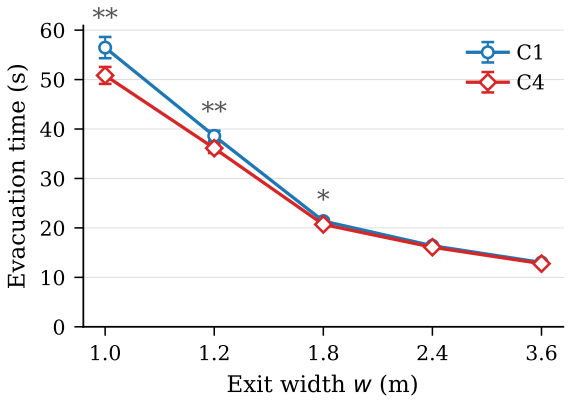

Figure 5: Evacuation time vs. exit width for C1 and C4. Error bars: 95% CIs, $n = 25$.

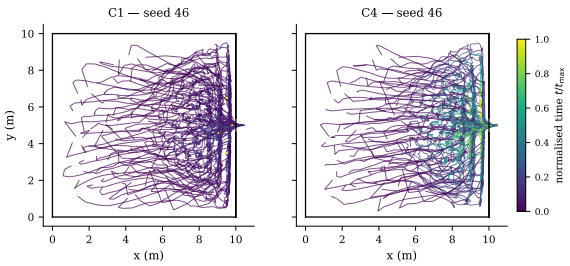

Figure 6: Agent trajectories at $w = 0.8$ m (100 agents, seed 46). **Left:** C1 forms a stable arching deadlock at the exit throat. **Right:** C4 destabilises the arch and evacuates all 100 agents. Colour: normalised time.

check. Parameters are not harmonised across the two simulators (desired speed, agent radius, and relaxation time all differ between tool defaults and our Table 1), so this comparison does not constitute an external quantitative validation; it asks only whether the qualitative ordering of steering paradigms reproduces in an independent implementation. JuPedSim's CollisionFreeSpeed (CFS) model evacuates in 33.6±1.2 s, 34% faster than our C4; JuPedSim SocialForce (SF), the closest analogue to our C1, exits only 31.3±2.3 of 50 agents in 60 s (0/25 complete) compared to 50/50 for our C1. The parameter gap produces opposite-direction deviations, illustrating that identical parameter differences can either hurt or help evacuation depending on the steering paradigm in use. The qualitative ordering (velocity-based models evacuate faster than force-based models at moderate bottleneck widths) is consistent across both frameworks, supporting the interpretation that the ordering is a property of the steering paradigms themselves rather than an implementation artefact. JuPedSim's C++ core runs approximately 39× faster than our pure-Python implementation (Table 7); this is a well-understood performance gap and does not affect the diagnostic conclusions of the paper.

Table 7: External simulator comparison at $w = 1.0$ m bottleneck, 50 agents, $n = 25$ seeds. JuPedSim uses tool-default parameters (not tuned to match ours). JuPedSim SF did not complete evacuation in any seed within 60 s; agents exited is reported instead of evacuation time.

Tool	**Evac. time (s)**	**Agents exited**	**Contact-overlaps**	**Runtime (s)**
Ours C1 (SFM)	56.5 ± 5.5	50/50	42 ± 17	17.5
Ours C4 (hybrid)	50.8 ± 4.4	50/50	29 ± 12	77.0
JuPedSim CFS	33.6 ± 1.2	50/50	—	0.4
JuPedSim SF	— (0/25 complete)	31.3 ± 2.3/50	—	0.6

4.8 Computational Performance

All timings were measured on Apple M1, 8 GB RAM, Python 3.13.12 (single core). Figure 7 shows per-step wall-time scaling. C1 rises from 2.0 ms at 50 agents to 744 ms at 1000 agents, consistent with the sparse $O(N{\cdot}k)$ KDTree pair computation that dominates force-based steering. C4 is dominated by ORCA's per-agent 2D linear program: 58.7 ms at 50 agents and 6 347 ms at 500 agents. ORCA introduces 29–36× overhead over SFM at the tested agent counts. The dominant hotspot is the ORCA LP solve (38% of C4 wall-time at 200 agents; 82% including the half-plane construction); C/C++ acceleration of the ORCA loop, or parallelisation across agents, would be required for real-time rates (≤ 33 ms/step at $\Delta t = 0.01$ s) beyond approximately 30 agents for C4. We did not profile beyond 500 agents for C4 owing to cost; linear extrapolation of the observed trend predicts ~ 13 s/step at 1 000 agents, which is consistent with the real-time-ineligible conclusion at larger populations.

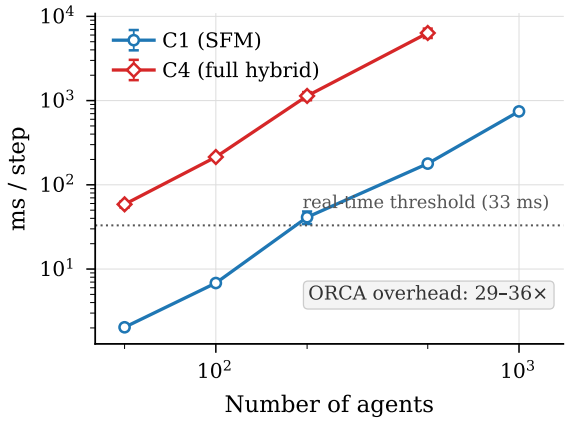

Figure 7: Per-step wall-time scaling (log–log). C1 measured to 1 000 agents; C4 to 500. ORCA introduces 29–36× overhead over SFM. Hardware: Apple M1, Python 3.13.12. Dotted line: real-time threshold (33 ms/step at $\Delta t = 0.01$ s).

5 DISCUSSION

Why each component matters, and why each can also harm. The ablation reveals that TTC and ORCA serve distinct, complementary roles, and that each can be harmful in the wrong regime. TTC-enabled configurations reduce contact-overlaps by 30% (IRR $= 0.695$, $p_{\mathrm{HS}} < 0.001$), concentrated upstream in the queue-formation zone, and at 90° crossings TTC's predictive evasion dissolves force-balance standoffs to approximately triple throughput; yet TTC alone does not break symmetric arches at narrow exits (C2: 0/25 at $w = 0.8$ m). ORCA-enabled configurations resolve the arch in 52–56% of trials while neither C1 nor C2 resolves it at all; yet ORCA alone depresses counterflow throughput (C3 coef. -5.0 exits in 60 s relative to C1) because its collision-free velocity selection makes agents yield to counterflow neighbours and produces gridlock. The full hybrid (C4) recovers throughput because TTC's contact-overlap reduction prevents the gridlock that pure ORCA yielding would otherwise precipitate. The σ-sweep locates the noise threshold for deadlock resolution at $\sigma_{50} = 0.049$ m/s (95% CI $[0.044, 0.053]$); passive Gaussian noise at $\sigma \approx \sigma_{50}$ recovers ORCA's effect within sampling resolution, and within $n = 25$ the residual contribution of ORCA's velocity-space optimisation beyond symmetry-breaking is not separable from noise. As a Δt-independent supplement, mean contact-event counts at $w = 1.0$ m are 18.5 for C1 versus 15.3 for C4, preserving the direction of the overlap-count result. Both negative findings — TTC harming deadlock resolution, ORCA harming counterflow — show that naïve always-on superposition of steering paradigms can be *worse* than any single paradigm. This is the motivation for the density-gated architecture: by weighting the one paradigm whose applicability is regime-dependent, the framework activates it where it helps and fades it where it would hurt, while the always-on paradigms remain available at every density.

Theoretical contribution. Beyond the specific numerical findings, the paper makes a methodological point that generalises beyond the SFM/TTC/ORCA triad. Zone-decomposed attribution — separating the upstream queue from the exit throat and from the downstream flow — lets us identify *which* part of a geometry a given paradigm's failure inhabits, and therefore which of a hybrid's components is doing the observed work. In our bottleneck results, 96% of the contact-overlap reduction is upstream; reporting only the aggregate 31% would have obscured the mechanism entirely. The same logic suggests that aggregate evacuation times conflate upstream queue dynamics with throat dynamics, and that progress in hybrid pedestrian models will benefit from the routine inclusion of spatial decompositions alongside global metrics.

Implications for practitioners. Three concrete, operational recommendations follow from the ablation. *For narrow-exit bottlenecks* ($w \leq 1.0$ m), plain SFM is categorically unsafe: $0/25$ evacuations at w=0.8 m is a qualitatively different prediction from $14/25$, and downstream safety analyses built on plain SFM at narrow apertures must either add ORCA-style velocity selection or, equivalently within our sampling resolution, explicit isotropic velocity noise at $\sigma \geq 0.05$ m/s. *For 90° crossings*, TTC anticipation is essential: both C1 and C3 reach force-balance standoffs that neither SFM nor ORCA alone dissolves. *For bidirectional counterflow*, avoid ORCA-only configurations: reciprocal yielding produces gridlock that TTC's anticipatory avoidance prevents in the full blend. More broadly, the 26–43% underestimation of empirical bottleneck flow after FD-only calibration is a consistent bias that no choice within the steering triad closes; applied users should expect a geometry-specific calibration step before trusting quantitative flow predictions. The blend functions as a mitigation probe rather than a production controller, and we do not recommend its deployment in safety-critical settings without the further validation enumerated in §6.

Oracle baseline. Across eight scenario–metric pairs, the full blend matches or slightly exceeds the best single paradigm in seven, without runtime scenario classification. The single $+1.4\%$ underperformance is at $w = 3.6$ m, where throughput is saturated and the oracle paradigm therefore has little margin to beat; no scenario shows C4 significantly worse than the best single paradigm.

Phenomenological calibration. The 2-parameter $(\gamma, \rho_{\max})$ Nelder–Mead fit reduces in-sample FD RMSE by 57.7%. Because Figure 1 shows the FD is identical across C1–C4, this fit does *not* validate the steering-paradigm comparison; it is reported as a phenomenological alignment of the speed–density curve rather than as evidence for the blend. Restricting the fit to $(\gamma, \rho_{\max})$ with A, B held at textbook values is a deliberate identifiability choice: a full 4-parameter refit with multi-restart global search is future work.

Scope, limitations, and what is not claimed. The additive blend is a mitigation probe rather than a deployable controller; we make no production-readiness claim, and calibration and ablation results are validated at the FD and flow-rate level but not at the steering-trajectory level. Replication counts of $n = 25$ reflect the ORCA LP compute cost (§4.8) and, together with the MDE statement of §4, bound the effect sizes this paper can resolve. The held-out bottleneck bias of -26% to -43% indicates that the framework does not generalise quantitatively across geometries, even when qualitative orderings are preserved; the per-paradigm OOD decomposition (Table 4) covers 2 of 5 held-out widths owing to the same compute constraint. The density-adaptive gate is not exercised in its intended transition regime in our scenarios (§4.4), so it behaves as a near-fixed blend in practice; a sensitivity sweep over (ρ_0, k) and high-density benchmarks are explicit future targets. The scaling analysis characterises computational cost on a single workstation and does not demonstrate real-time deployment; the external simulator comparison uses tool-default parameters rather than a harmonised cross-simulator calibration; and the sigmoid transitions are not hysteretic, which we would recommend for deployments at densities sustained near ρ_0.

6 CONCLUSIONS

We presented a diagnostic zone-decomposed ablation of three pedestrian steering paradigms (SFM, TTC, ORCA), identifying two paradigm-specific failure modes, a mechanism-level deadlock study via the $C_1+\varepsilon$ control, and a density-modulated additive blend as a mitigation probe. Held-out FZJ bottleneck flow rates are systematically underestimated by 26–43%, indicating that FD-only calibration generalises within-corridor but not across-geometry. FD calibration aligns the speed–density relation (57.7% in-sample RMSE reduction) but does not validate steering — the FD is identical across C1–C4. The noise-threshold control ($\sigma_{50} \approx 0.049$ m/s, 95% CI $[0.044, 0.053]$) shows that at $n = 25$ passive Gaussian noise is indistinguishable from ORCA's LP-based velocity selection as a deadlock-breaking mechanism. Taken together, these findings support a methodological reorientation: in pedestrian simulation, zone-decomposed ablation and honest OOD reporting are more informative than aggregate benchmarks, and blends that gate only the paradigm whose applicability is regime-dependent are a natural mitigation architecture. Code, data, and reproduction scripts are released with the paper.

Future work targets (i) a sensitivity analysis over the density-gate and steering hyperparameters (ρ_0, k, A, B); (ii) trajectory-level validation on additional geometries (ETH–UCY and further FZJ scenarios); (iii) higher-density scenarios that genuinely exercise the adaptive transition region; (iv) a larger-n σ-sweep to separate passive-noise symmetry-breaking from ORCA's velocity-space optimisation; (v) C/C++ acceleration of the ORCA LP to unlock real-time scale; and (vi) a full 4-parameter multi-restart calibration of $(\gamma, \rho_{\max}, A, B)$. A natural longer-term direction for the mitigation-probe architecture is evaluation as a component of a digital-twin pipeline once trajectory-level validation is in place; that evaluation is out of scope here.

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A REPRODUCIBILITY

A.1 Environment. Apple M1, 8 GB RAM, macOS 26.3, Python 3.13.12 (single CPU for serial sims; `multiprocessing.Pool(10)` for cross-bin calibration). Packages: numpy 2.4.4, scipy 1.17.1, shapely 2.1.2, statsmodels 0.14, lifelines 0.28. Commit hash: `[anonymized for review]`.

A.2 Geometry. Corridor 18×5 m (periodic-x, FD calibration); bottleneck 10×10 m with exit width $w \in \{0.8, 1.0, 1.2, 1.8, 2.4, 3.6\}$ m; bidirectional corridor 20×3.6 m; 90° crossing (two 20 m perpendicular corridors 3.6 m wide). Wall thickness 0.1 m. Agents: v_0=1.34 m/s, r=0.25 m, m=80 kg, τ=0.5 s.

A.3 Hyperparameters. Calibrated: γ^*=0.8327, $\rho^*_{\max}$=5.977 (Table 1). Textbook-fixed (Helbing et al., 2000; Karamouzas et al., 2014; van den Berg et al., 2011): A=2000 N, B=0.08 m, k=1.2×10^5 kg/s^2, κ=2.4×10^5 kg/(m·s), k_{ttc}=1.5, τ_0=3.0 s, τ_h=5.0 s. Implementation: τ_{ORCA}=0.5 s, R=3.0 m (KDTree), ρ_0=4.0 ped/m^2, gate slope k=2.0, Δt=0.01 s (Euler).

A.4 Voronoi density. Shapely tessellation clipped to scenario boundary; Mullick mirror-points (4 reflections per boundary agent) eliminate unbounded cells. $\rho_i = 1/\text{area}(\text{cell}_i)$.

A.5 Calibration. Nelder–Mead, single-start, 2 free params ($\gamma, \rho_{\max}$), A/B fixed. Bounds $\gamma \in [0.3, 3.0]$, $\rho_{\max} \in [3.0, 7.0]$; $x_{\text{atol}}=10^{-3}$, $f_{\text{atol}}=10^{-4}$; seeds-per-eval = 3 (seeds 42–44); 10 FD density bins in $[0.5, 5.0]$ ped/m^2. Terminated at evaluation 53 (simplex oscillating ±0.003 m/s at the seed-noise floor). Baseline RMSE 0.23509 → 0.09945 m/s.

A.6 Statistics. NB GLM via statsmodels (log link, MoM α, duration offset for per-run rates); Cox PH via lifelines; LMM via statsmodels MixedLM (random intercept per seed); Holm–Šidák correction within per-scenario comparison families; all seeds retained (no exclusions). Cox PH uses the Efron tie-handling. LMM normality diagnostics (QQ plots of level-1 residuals) are in the artefact repository.

A.7 Data. FZJ unidirectional corridor: 4 776 (ρ, v) points aggregated from 952 trajectories (`data/fzj/unidirectional/`). FZJ bottleneck (held-out OOD): per-width flow rates (`data/fzj/bottleneck/`).

A.8 Reproduction. Calibration: `python scripts/session06_phaseD_alight.py`. Bottleneck: `python scripts/run_experiments.py --scenario bottleneck --seeds 42-66`. σ-sweep: `python scripts/sigma_sweep.py`. Zonal decomposition: `python analysis/zonal_decomposition.py`. External sim: `python new_experiments/external_simulator/jupedsim_runner.py`. Figures: `python scripts/regenerate_all_figures.py`. Tables: `python scripts/regenerate_tables.py`.

A.9 Artefacts. Key files (sha256 prefixes): `results_new/sigma_sweep.csv` (ce706dbf...) → Fig. 4, §4.6; `calibration_alight_result.json` → Table 1, §4.2; `results_analysis/zonal_collisions.csv` → Fig. 3, Table 5; `ood_validation.csv` → §4.3, Table 3; `oracle_baseline.md` → §4.5.